

Secure Image & File Steganography Tool

Introduction

Steganography is the technique of hiding secret information inside digital media such as images, audio, or video files. Unlike encryption, which protects the content of a message, steganography conceals the very existence of the message. This project, Secure Image & File Steganography Tool, provides a graphical user interface (GUI) that allows users to securely hide and extract text messages or files within images. The application supports drag-and-drop functionality, image preview, encryption, and extraction features, making it a practical solution for secure data hiding.

Abstract

The Secure Image & File Steganography Tool is a Python-based desktop application developed to hide confidential text messages and files inside image files using Least Significant Bit (LSB) steganography. Before embedding, the data is encrypted to enhance security. The application supports common image formats such as PNG and BMP and provides a user-friendly GUI built with Tkinter. Users can drag and drop images, browse files, hide messages, hide files, and extract hidden data with ease. This project demonstrates the practical implementation of steganography and encryption techniques for secure communication and data protection.

Tools Used

- Python 3
- Tkinter (GUI Development)
- TkinterDnD2 (Drag-and-Drop Support)
- Pillow (PIL) for Image Processing
- Cryptography Library for Encryption and Decryption
- Base64 Encoding for File Handling
- Git & GitHub for Version Control

Steps Involved in Building the Project

1. Designed the project architecture and user interface layout.
2. Implemented image selection and drag-and-drop functionality.
3. Developed image preview support using the Pillow library.
4. Implemented LSB-based steganography for embedding secret data inside images.
5. Added encryption and decryption using the Cryptography library.
6. Developed functionality to hide and extract text messages.
7. Implemented file hiding and extraction using Base64 encoding.

8. Added support for PNG and BMP image formats.
9. Integrated error handling and validation for user inputs.
10. Tested all functionalities and optimized the GUI for better user experience.
11. Managed source code using Git and published the project on GitHub.

Conclusion

The Secure Image & File Steganography Tool successfully demonstrates how sensitive information can be securely hidden inside images using steganography and encryption techniques. The project provides both message and file hiding capabilities through an intuitive graphical interface. Features such as drag-and-drop support, image preview, encryption, and file extraction make the application practical and user-friendly. This project helped in understanding image processing, data security, GUI development, and software version control while delivering a functional cybersecurity-related application.